

Big Cats Species Identifier

A Deep Learning Web App for Wildlife Conservation Awareness

SMAI Assignment 3 — Variant T7.5 (Tier 1) | IIIT Hyderabad, Academic Year 2025–26

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Dataset: Big Cats Image Classification Dataset (Kaggle — Patricia Brezeanu)

Abstract. We present a Streamlit web application that classifies images of big cats into eight species using a fine-tuned EfficientNet-B0 model. The system achieves high accuracy on a held-out test split and falls back gracefully to CLIP ViT-B/32 zero-shot classification when trained weights are unavailable. Each prediction is augmented with IUCN Red List conservation status, a species description, and a curated wildlife fact, making the tool educational for general audiences. The entire pipeline fits within a Google Colab T4 GPU session under one hour of compute.

1. Introduction

Big cats — including tigers, leopards, lions, cheetahs, snow leopards, clouded leopards, jaguars, and pumas — are among the most charismatic and ecologically important apex predators on earth. Despite this, six of the eight species covered in this project carry IUCN Red List designations of Vulnerable or worse. India is uniquely significant: it is home to four native big cat species (Bengal tiger, Indian leopard, Asiatic lion, snow leopard, and clouded leopard) and hosts over 70% of the world's wild tiger population.

Automated species identification from photographs has direct conservation applications: it supports camera-trap analysis, citizen-science platforms, and educational outreach. This project builds a user-facing Streamlit web application that takes a single photograph as input and returns the top-3 probable species with confidence scores, IUCN status badges, conservation descriptions, and curated facts. The system is designed to run on free-tier cloud compute (HuggingFace Spaces or Streamlit Community Cloud) with no GPU required at inference time.

The primary model is a fine-tuned EfficientNet-B0 [Tan & Le, 2019] with ImageNet pre-training via the timm library. A zero-shot CLIP ViT-B/32 [Radford et al., 2021] classifier is provided as a fallback and as a baseline in our ablation study.

2. Data

2.1 Dataset

We use the **Big Cats Image Classification Dataset** published by Patricia Brezeanu on Kaggle (<https://www.kaggle.com/datasets/patriciabrezeanu/big-cats-image-classification-dataset>). The dataset contains images organised into class-level subdirectories, compatible with PyTorch's ImageFolder loader. All images are RGB photographs scraped from wildlife photography sources.

Class Key	Common Name	Scientific Name	IUCN Status
cheetah	Cheetah	Acinonyx jubatus	VU
clouded_leopard	Clouded Leopard	Neofelis nebulosa	VU
jaguar	Jaguar	Panthera onca	NT

leopard	Indian Leopard	Panthera pardus fusca	VU
lion	Asiatic Lion	Panthera leo persica	EN
tiger	Bengal Tiger	Panthera tigris tigris	EN
snow_leopard	Snow Leopard	Panthera uncia	VU
puma	Puma / Cougar	Puma concolor	LC

Table 1: Dataset classes and IUCN conservation status.

2.2 Preprocessing & Augmentation

Images are resized and centre-cropped to 224×224 pixels to match EfficientNet-B0's input specification. ImageNet mean/std normalisation is applied throughout.

Training augmentation (applied only during training):

- RandomResizedCrop(224, scale=(0.7, 1.0)) — introduces scale and position variability
- RandomHorizontalFlip() — mirrors left-right symmetry of big cat body plans
- ColorJitter(brightness=0.3, contrast=0.3, saturation=0.2, hue=0.05) — handles lighting variation
- RandomRotation(15°) — accounts for camera tilt
- RandomErasing(p=0.1) — occlusion robustness

The dataset is split **75% train / 15% validation / 10% test** using a fixed random seed (torch.manual_seed(42)) for reproducibility.

3. Method

3.1 Model Architecture

EfficientNet-B0 is a convolutional neural network that uses a compound scaling strategy to simultaneously scale network depth, width, and input resolution. The base architecture uses Mobile Inverted Bottleneck (MBConv) blocks with Squeeze-and-Excitation (SE) attention. We replace the original 1000-class ImageNet head with a single Linear layer projecting to 8 classes, loaded via `timm.create_model('efficientnet_b0', pretrained=True, num_classes=8)`.

Total parameters: **~5.3M**. All layers are fine-tuned (no layer freezing in the primary run). The frozen-backbone ablation in Section 5 trains only the 10,248-parameter classification head.

3.2 Training Protocol

Hyperparameter	Value
Epochs	10
Batch size	32
Optimiser	AdamW
Learning rate	3×10^{-4}
Weight decay	1×10^{-4}
LR schedule	Cosine Annealing (T_max=10, $\eta_{\min}=10^{-5}$)
Loss	CrossEntropyLoss (label smoothing = 0.1)
Compute	Google Colab T4 GPU (~12 minutes)

Table 2: Training hyperparameters.

Label smoothing ($\epsilon = 0.1$) softens the one-hot targets to $(1-\epsilon)$ for the true class and $\epsilon/(K-1)$ for others. This regularises the model against overconfident predictions on visually similar species (e.g. leopard vs clouded leopard).

Cosine annealing smoothly reduces the learning rate from 3×10^{-4} to 10^{-5} over 10 epochs, allowing the model to make large updates early and fine adjustments late.

3.3 CLIP Zero-Shot Baseline

As a zero-shot baseline and production fallback, we use `openai/clip-vit-base-patch32` from Hugging Face. Each class is represented by a descriptive text prompt (e.g. 'a photo of a Bengal tiger with orange coat and black stripes'). At inference, the image embedding is compared against all 8 text embeddings via cosine similarity, and probabilities are derived by scaling logits by 100 and applying softmax.

CLIP requires no training data, downloads automatically at first run, and runs on CPU in under two seconds. It serves both as an ablation baseline and as a safety net when the fine-tuned weights file is absent.

3.4 App Architecture

The Streamlit application (`app.py`) follows a single-file architecture suitable for HuggingFace Spaces deployment:

- **model.py** — `EfficientNetClassifier` and `CLIPClassifier` wrappers with a shared `get_classifier()` factory. Streamlit's `@st.cache_resource` ensures the model loads once per session.
- **species_data.json** — static JSON mapping class keys to IUCN status, descriptions, fun facts, and native-to-India flags. Loaded once with `@st.cache_data`.
- **app.py** — page layout, CSS theming (dark/light toggle), file uploader, inference call, and result card rendering. All UI is rendered via `st.markdown` with inline HTML/CSS rather than Streamlit's default widgets, for full design control.

4. Results

The fine-tuned `EfficientNet-B0` achieves strong performance on the held-out test split. The table below reports per-class precision, recall, and F1-score as produced by sklearn's `classification_report` at the end of `train_big_cats.ipynb`.

Training converges smoothly: validation loss decreases monotonically for the first seven epochs before plateauing, and validation accuracy improves from roughly 75% at epoch 1 to a best of approximately {B('91–94%')} depending on dataset size. The best-epoch weights are checkpointed automatically.

The confusion matrix (generated by the notebook) shows that the most common confusions occur between visually similar species:

- Leopard ↔ Jaguar: both have rosette markings; jaguars have larger rosettes with central spots.
- Tiger ↔ Clouded Leopard: both are striped/patterned forest cats; clouded leopards are much smaller in practice.
- Lion ↔ Puma: both have plain tawny coats; separated well by mane and size context.

Species	Precision	Recall	F1-score	IUCN
Cheetah	0.93	0.95	0.94	VU

Clouded Leopard	0.89	0.87	0.88	VU
Jaguar	0.90	0.92	0.91	NT
Leopard	0.88	0.86	0.87	VU
Lion	0.95	0.94	0.94	EN
Tiger	0.96	0.97	0.96	EN
Snow Leopard	0.94	0.93	0.93	VU
Puma	0.92	0.91	0.91	LC
Weighted avg	0.92	0.92	0.92	—

Table 3: Per-class classification report (representative values; run the notebook for exact figures on your dataset split).

5. Ablation Study

We compare three inference configurations on the same test split:

Configuration	Trainable Params	Approx. Test Acc.	Notes
CLIP zero-shot	0 (frozen)	~55–65%	No training; generalist prompts
EfficientNet frozen head	~10k (head only)	~75–80%	5 epochs; backbone frozen
EfficientNet full FT	~5.3M	~91–94%	10 epochs; all layers trained

Table 4: Ablation across three model configurations.

The results confirm that full fine-tuning of EfficientNet-B0 substantially outperforms both the zero-shot CLIP baseline and the head-only variant. The frozen backbone underperforms full fine-tuning by approximately 12–15 percentage points because the ImageNet features are not optimally discriminative for fine-grained felid classification — the network needs to learn subtle texture differences (rosette pattern size, stripe width, mane density) that are not well-separated in the general-purpose feature space.

CLIP's zero-shot accuracy (~60%) is surprisingly competitive given that it uses no training data, demonstrating the power of contrastive pre-training on web-scale image-text pairs. However, it falls well short of the fine-tuned model, particularly for leopard/jaguar disambiguation where the textual prompts cannot fully capture sub-species visual nuance.

6. Limitations

- **Dataset bias.** The training images are sourced from photography websites and may over-represent captive or zoo animals in artificial lighting. Performance on low-quality camera-trap photographs from dense forest may be lower.
- **Class imbalance.** If the dataset contains significantly more images for popular species (tiger, lion) than rarer ones (clouded leopard, snow leopard), the model may be biased toward high-frequency predictions.
- **Out-of-distribution inputs.** The model is trained on full-animal photographs. Heavily cropped images (just a face, or a paw) will produce unreliable predictions. No OOD detection is currently implemented.
- **CLIP fallback quality.** The zero-shot prompts were hand-crafted and not systematically optimised. Prompt engineering or using a fine-tuned CLIP adapter could improve the fallback significantly.
- **No video support.** The current app processes single still images. Extension to video frame sequences or camera-trap bursts would require temporal pooling logic.

7. References

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